



Advanced Dynamic Folder for MX Creative Console

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Goal

To expand the current profiles in Logitech Options+.

Make the console smarter. Change contents according to user's behavior in the app.

The contents on the console will change when user select a element or executing specific actions, not just only opening apps.

Office Addin(Excel) selecting a cell would generate copy and paste functions, cut functions, etc commonly used functions.

Features

Our project consists of two parts the plugin and a demo add in.

Add in(Excel) hooks onto plugin through a websocket.

Plugin receives application data through the websocket.

Plugin changes button function depending on user behavior such as selecting buttons.



Methods

From Logitech Actions SDK, we used dynamic folders so that we could adjust button functions quickly.

Office JavaScript API, used to gather user actions through API

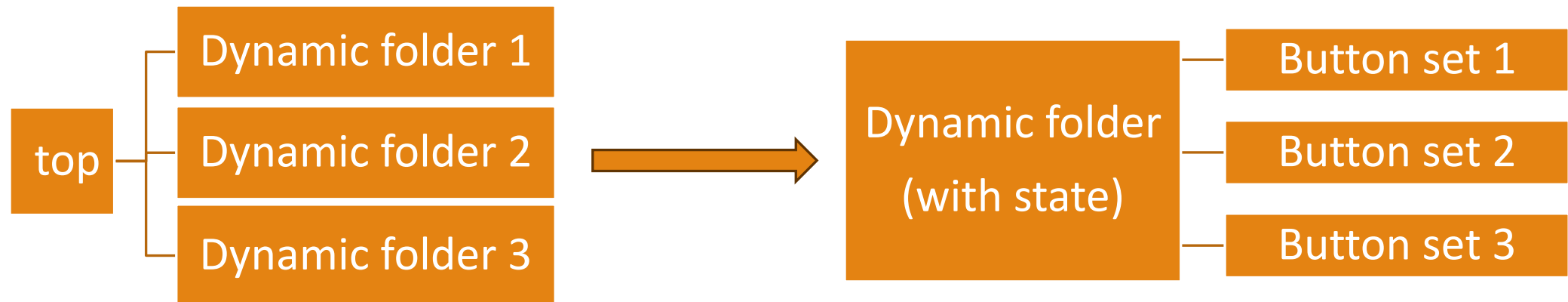
Websocket local server, used to receive API data then send to our console plugin.

Problems we faced (1)

Unable to simulate physical button press through software. Thus we solved this by changing button functions in our dynamic folder.

How we solved:

We originally want to use multiple dynamic folders to hold all our generated button functions, but discovered that we can implement just one and instead use internal state variables dynamically buffer button sets according to state.

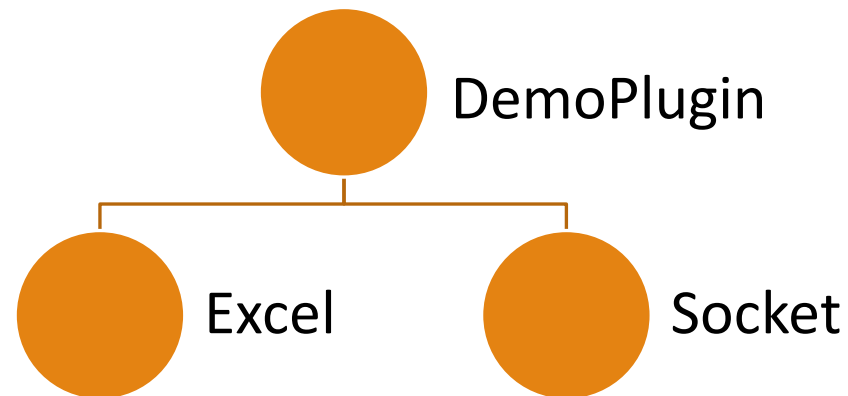


Problems we faced (2)

We tried to change the contents of the dynamic folder through our websocket, but we currently can't due to a misconfigured data path. Where we might have sent the data to the wrong instance, and the actual button remained unchanged.

How we solved:

After restructuring our data types and class objects, we found the missing link and allowed our button to receive the appropriate state.



Future outlook

As long as a software API supports websocket, we would be able to add the software into the support list, thus high expandability.

Example, all Microsoft office applications would be able to be quickly added into our current demo.

Leave some functions to be recommended not based on user actions, but through the use of AI instead.